Lengthy Derivations in Sect. 6.3.1

Basic Integrals (1)

2.26 Forms containing $\sqrt{a+bx+cx^2}$ and integral powers of x

Notation: $R = a + bx + cx^2, \quad \Delta = 4ac - b^2$

2.261[11] For $n = -1$

$$\int \frac{dx}{\sqrt{R}} = \frac{1}{\sqrt{c}} \ln\left(\frac{2\sqrt{cR} + 2cx + b}{\sqrt{\Delta}}\right) \qquad [c > 0]$$

$$= \frac{1}{\sqrt{c}} \operatorname{arcsinh}\left(\frac{2cx+b}{\sqrt{\Delta}}\right) \qquad [c > 0, \quad \Delta > 0]$$

$$= \frac{1}{\sqrt{c}} \ln(2cx + b) \qquad [c > 0, \quad \Delta = 0]$$

$$= \frac{-1}{\sqrt{-c}} \arcsin\left(\frac{2cx+b}{\sqrt{-\Delta}}\right) \qquad [c < 0, \quad \Delta < 0]$$

5. $$\int \frac{dx}{\sqrt{R^3}} = \frac{2(2cx+b)}{\Delta\sqrt{R}}$$

6. $$\int \frac{x\,dx}{\sqrt{R^3}} = -\frac{2(2a+bx)}{\Delta\sqrt{R}}$$

7. $$\int \frac{x^2\,dx}{\sqrt{R^3}} = -\frac{\left(\Delta - b^2\right)x - 2ab}{c\Delta\sqrt{R}} + \frac{1}{c}\int\frac{dx}{\sqrt{R}} \qquad \text{(see } \mathbf{2.261}\text{)}$$

8. $$\int \frac{x^3\,dx}{\sqrt{R^3}} = \frac{c\Delta x^2 + b\left(10ac - 3b^2\right)x + a\left(8ac - 3b^2\right)}{c^2\Delta\sqrt{R}} - \frac{3b}{2c^2}\int\frac{dx}{\sqrt{R}}$$

(see **2.261**)

$$Q = 1 + x^2 - 2x\cos\psi$$

$$a = 1, \qquad b = -2\cos\psi, \qquad c = 1, \qquad \Delta = 4(1 - \cos^2\psi) = 4\sin^2\psi$$

$$\int \frac{x^2 dx}{Q^{3/2}} = -\frac{4(1 - \cos^2\psi - \cos^2\psi)x + 4\cos\psi}{4(1-\cos^2\psi)Q^{\frac{1}{2}}} + \ln\left(\frac{2Q^{\frac{1}{2}} + 2x - 2\cos\psi}{2\sin\psi}\right) + C$$

$$\int \frac{x^2 dx}{Q^{3/2}} = -\frac{(1 - 2\cos^2\psi)x + \cos\psi}{(1-\cos^2\psi)Q^{\frac{1}{2}}} + \ln\left(\frac{Q^{\frac{1}{2}} + x - \cos\psi}{\sin\psi}\right) + C$$

$$\int \frac{x^2 dx}{Q^{3/2}} = -\frac{(1 - 2\cos^2\psi)x + \cos\psi}{(1-\cos^2\psi)Q^{\frac{1}{2}}} + \ln\left(\frac{Q^{\frac{1}{2}} + x - \cos\psi}{\sin\psi}\right) + C$$

Basic Integrals (2)

$$u = \sqrt{a + cx^2}$$

$$I_1 = \frac{1}{\sqrt{c}} \ln\left(x\sqrt{c} + u\right)$$

$$\int \frac{dx}{u} = I_1$$

$$\int \frac{dx}{u^3} = \frac{1}{a}\frac{x}{u}$$

$$\int \frac{dx}{u^{2n+1}} = \frac{1}{a^n} \sum_{k=0}^{n-1} \frac{(-1)^k}{2k+1} \binom{n-1}{k} \frac{c^k x^{2k+1}}{u^{2k+1}}$$

$$\int \frac{x\,dx}{u^{2n+1}} = -\frac{1}{(2n-1)cu^{2n-1}}$$

$$a = 2 - 2\cos\psi\,, \qquad c = 1$$

$$\int \frac{d\delta x}{(2 - 2\cos\psi + \delta x^2)^{1/2}} = \ln\left[\delta x + (2 - 2\cos\psi + \delta x^2)^{1/2}\right] + C$$

$$\int \frac{\delta x\, d\delta x}{(2 - 2\cos\psi + \delta x^2)^{1/2}} = (2 - 2\cos\psi + \delta x^2)^{1/2} + C$$

$$\int \frac{\delta x\, d\delta x}{(2 - 2\cos\psi + \delta x^2)^{3/2}} = -\frac{1}{(2 - 2\cos\psi + \delta x^2)^{\frac{1}{2}}} + C$$

$$\int \frac{\delta x\, d\delta x}{(2 - 2\cos\psi + \delta x^2)^{3/2}} = -\frac{1}{(2 - 2\cos\psi + \delta x^2)^{1/2}} + C$$

$$\int \frac{d\delta x}{(2 - 2\cos\psi + \delta x^2)^{3/2}} = \frac{\delta x}{(2 - 2\cos\psi)(2 - 2\cos\psi + \delta x^2)^{1/2}} + C$$

$$\int \frac{\delta x\, d\delta x}{(2 - 2\cos\psi + \delta x^2)^{5/2}} = -\frac{1}{3(2 - 2\cos\psi + \delta x^2)^{3/2}} + C$$

Eq. (6.61)

(6.61) is straightforward with integral formula

$$\int \frac{x^2 dx}{(1+x^2-2x\,cos\,\psi)^{1/2}}$$
$$= \left(\frac{x}{2}+\frac{3\,cos\,\psi}{2}\right)(1+x^2-2x\,cos\,\psi)^{1/2}$$
$$+\left(\frac{3\,cos^2\psi}{2}-\frac{1}{2}\right) ln\left[(1+x^2-2x\,cos\,\psi)^{1/2}+x-cos\,\psi\right]+C$$

This formula is from the following formula copied from page 16 of this document:

$$\int \frac{x^2 dx}{\sqrt{R}} = \left(\frac{x}{2}+\frac{3\cos\Theta}{2}\right)(1+x^2-2x\cos\Theta)^{\frac{1}{2}}$$
$$+\left(\frac{3\cos^2\Theta}{2}-\frac{1}{2}\right)\ln\left[(1+x^2-2x\cos\Theta)^{\frac{1}{2}}+x-\cos\Theta\right]+C$$

Eq. (6.65)—(6.67)

$$f(x) = (x+3\,cos\,\psi)(1+x^2-2x\,cos\,\psi)^{1/2}$$
$$+(3\,cos^2\psi-1)\,ln\left[(1+x^2-2x\,cos\,\psi)^{1/2}+x-cos\,\psi\right]$$

$$f'(x) = (1+x^2-2x\,cos\,\psi)^{1/2}+\frac{(x+3\,cos\,\psi)(x-cos\,\psi)}{(1+x^2-2x\,cos\,\psi)^{1/2}}$$
$$+\frac{(3\,cos^2\psi-1)\left[\frac{x-cos\,\psi}{(1+x^2-2x\,cos\,\psi)^{1/2}}+1\right]}{(1+x^2-2x\,cos\,\psi)^{1/2}+x-cos\,\psi}$$
$$=(1+x^2-2x\,cos\,\psi)^{1/2}+\frac{x^2-x\,cos\,\psi+3x\,cos\,\psi-3\,cos^2\psi}{(1+x^2-2x\,cos\,\psi)^{1/2}}$$
$$+\frac{3\,cos^2\psi-1}{(1+x^2-2x\,cos\,\psi)^{1/2}}$$
$$=(1+x^2-2x\,cos\,\psi)^{1/2}+\frac{x^2+2x\,cos\,\psi-3\,cos^2\psi+(3\,cos^2\psi-1)}{(1+x^2-2x\,cos\,\psi)^{1/2}}$$
$$=(1+x^2-2x\,cos\,\psi)^{1/2}+\frac{x^2+2x\,cos\,\psi-1}{(1+x^2-2x\,cos\,\psi)^{1/2}}$$
$$=(1+x^2-2x\,cos\,\psi)^{1/2}+\frac{2x^2+2x\,cos\,\psi-1-x^2}{(1+x^2-2x\,cos\,\psi)^{1/2}}$$
$$=(1+x^2-2x\,cos\,\psi)^{1/2}+\frac{2x\,cos\,\psi-1-x^2}{(1+x^2-2x\,cos\,\psi)^{1/2}}+\frac{2x^2}{(1+x^2-2x\,cos\,\psi)^{1/2}}$$
$$=\frac{2x^2}{(1+x^2-2x\,cos\,\psi)^{1/2}}$$

$$f''(x) = \frac{4x}{(1 + x^2 - 2x\, cos\, \psi)^{\frac{1}{2}}} - \frac{2x^2(x - cos\, \psi)}{(1 + x^2 - 2x\, cos\, \psi)^{\frac{3}{2}}}$$

$$= \frac{4x(1 + x^2 - 2x\, cos\, \psi) - 2x^2(x - cos\, \psi)}{(1 + x^2 - 2x\, cos\, \psi)^{\frac{3}{2}}}$$

$$= \frac{4x + 4x^3 - 8x^2\, cos\, \psi - 2x^3 + 2x^2\, cos\, \psi}{(1 + x^2 - 2x\, cos\, \psi)^{3/2}}$$

$$= \frac{4x + 2x^3 - 6x^2\, cos\, \psi}{(1 + x^2 - 2x\, cos\, \psi)^{\frac{3}{2}}}$$

$$f'''(x) = \frac{4 + 6x^2 - 12x \cos \psi}{(1 + x^2 - 2x\, cos\, \psi)^{3/2}} - \frac{3(4x + 2x^3 - 6x^2\, cos\, \psi)(2x - 2\cos \psi)}{2(1 + x^2 - 2x\, cos\, \psi)^{\frac{5}{2}}}$$

$$= \frac{4 + 6x^2 - 12x \cos \psi}{(1 + x^2 - 2x\, cos\, \psi)^{\frac{3}{2}}} - \frac{3(3x + 3x^3 - 6x^2\, cos\, \psi + x - x^3)(2x - 2\cos \psi)}{2(1 + x^2 - 2x\, cos\, \psi)^{\frac{5}{2}}}$$

$$= \frac{4 + 6x^2 - 12x \cos \psi}{(1 + x^2 - 2x\, cos\, \psi)^{\frac{3}{2}}}$$

$$- \frac{3(3x + 3x^3 - 6x^2\, cos\, \psi)(2x - 2\cos \psi) + 3(x - x^3)(2x - 2\cos \psi)}{2(1 + x^2 - 2x\, cos\, \psi)^{\frac{5}{2}}}$$

$$= \frac{4 + 6x^2 - 12x \cos \psi}{(1 + x^2 - 2x\, cos\, \psi)^{\frac{3}{2}}} - \frac{3(3x + 3x^3 - 6x^2\, cos\, \psi)(2x - 2\cos \psi)}{2(1 + x^2 - 2x\, cos\, \psi)^{\frac{5}{2}}}$$

$$- \frac{3(x - x^3)(2x - 2\cos \psi)}{2(1 + x^2 - 2x\, cos\, \psi)^{\frac{5}{2}}}$$

$$= \frac{4 + 6x^2 - 12x \cos \psi}{(1 + x^2 - 2x\, cos\, \psi)^{\frac{3}{2}}} - \frac{9x(x - \cos \psi)}{(1 + x^2 - 2x\, cos\, \psi)^{\frac{3}{2}}} - \frac{3(x - x^3)(2x - 2\cos \psi)}{2(1 + x^2 - 2x\, cos\, \psi)^{\frac{5}{2}}}$$

$$= \frac{4 + 6x^2 - 12x \cos \psi}{(1 + x^2 - 2x\, cos\, \psi)^{\frac{3}{2}}} - \frac{9x^2 - 9x \cos \psi}{(1 + x^2 - 2x\, cos\, \psi)^{\frac{3}{2}}} - \frac{(x - x^3)(2x - 2\cos \psi)}{(1 + x^2 - 2x\, cos\, \psi)^{\frac{5}{2}}}$$

$$= \frac{4 + 6x^2 - 12x \cos \psi - 9x^2 + 9x \cos \psi}{(1 + x^2 - 2x\, cos\, \psi)^{\frac{3}{2}}} - \frac{3(x - x^3)(2x - 2\cos \psi)}{2(1 + x^2 - 2x\, cos\, \psi)^{\frac{5}{2}}}$$

$$= \frac{4 - 3x^2 - 3x \cos \psi}{(1 + x^2 - 2x\, cos\, \psi)^{\frac{3}{2}}} - \frac{3(x - x^3)(2x - 2\cos \psi)}{2(1 + x^2 - 2x\, cos\, \psi)^{\frac{5}{2}}}$$

$$= \frac{8 - 6x^2 \ - 6x\cos\psi}{2(1 + x^2 - 2x\,cos\,\psi)^{\frac{3}{2}}} - \frac{3x(1 - x^2)(x - \cos\psi)}{(1 + x^2 - 2x\,cos\,\psi)^{\frac{5}{2}}}$$

$$= \frac{(3 + 3x^2 - 6x\cos\psi) + 5 - 9x^2}{2(1 + x^2 - 2x\,cos\,\psi)^{\frac{3}{2}}} - \frac{3x(1 - x^2)(x - \cos\psi)}{(1 + x^2 - 2x\,cos\,\psi)^{\frac{5}{2}}}$$

$$= \frac{3 + 3x^2 - 6x\cos\psi}{2(1 + x^2 - 2x\,cos\,\psi)^{\frac{3}{2}}} + \frac{5 - 9x^2}{2(1 + x^2 - 2x\,cos\,\psi)^{3/2}} - \frac{3x(1 - x^2)(x - \cos\psi)}{(1 + x^2 - 2x\,cos\,\psi)^{5/2}}$$

$$= \frac{3}{2(1 + x^2 - 2x\,cos\,\psi)^{\frac{1}{2}}} + \frac{5 - 9x^2}{2(1 + x^2 - 2x\,cos\,\psi)^{3/2}} - \frac{3x(1 - x^2)(x - \cos\psi)}{(1 + x^2 - 2x\,cos\,\psi)^{5/2}}$$

Eq. (6.85)

$$\int \frac{x^2 dx}{Q^{3/2}} = -\frac{4(1 - \cos^2\psi - \cos^2\psi)x + 4\cos\psi}{4(1 - \cos^2\psi)Q^{1/2}} + \ln\left(\frac{2Q^{1/2} + 2x - 2\cos\psi}{2\sin\psi}\right)$$

$$\int \frac{x^2 dx}{Q^{3/2}} = -\frac{(1 - 2\cos^2\psi)x + \cos\psi}{(1 - \cos^2\psi)Q^{1/2}} + \ln\left(\frac{Q^{1/2} + x - \cos\psi}{\sin\psi}\right)$$

$$\int \frac{x^2 dx}{Q^{3/2}} = -\frac{(1 - 2\cos^2\psi)x + \cos\psi}{(1 - \cos^2\psi)Q^{1/2}} + \ln\left(\frac{Q^{1/2} + x - \cos\psi}{\sin\psi}\right)$$

$$\int \frac{x^3 dx}{Q^{3/2}} = \frac{4(1 - \cos^2\psi)x^2 - 2\cos\psi\,(10 - 12\cos^2\psi)x + 8 - 12\cos^2\psi}{4(1 - \cos^2\psi)Q^{1/2}} + 3\cos\psi\ln\left(\frac{2Q^{1/2} + 2x - 2\cos\psi}{2\sin\psi}\right)$$

$$\int \frac{x^3 dx}{Q^{3/2}} = \frac{(1 - \cos^2\psi)x^2 - \cos\psi\,(5 - 6\cos^2\psi)x + 2 - 3\cos^2\psi}{(1 - \cos^2\psi)Q^{1/2}} + 3\cos\psi\ln\left(\frac{Q^{1/2} + x - \cos\psi}{\sin\psi}\right)$$

$$\int\left(\frac{x^2}{Q^{3/2}}-\cos\psi\frac{x^3}{Q^{3/2}}\right)dx$$

$$=-\frac{(1-2\cos^2\psi)x+\cos\psi}{(1-\cos^2\psi)Q^{\frac{1}{2}}}+\ln\left(\frac{Q^{\frac{1}{2}}+x-\cos\psi}{\sin\psi}\right)$$

$$-\cos\psi\frac{(1-\cos^2\psi)x^2-\cos\psi\,(5-6\cos^2\psi)x+2-3\cos^2\psi}{(1-\cos^2\psi)Q^{\frac{1}{2}}}$$

$$-3\cos^2\psi\ln\left(\frac{Q^{1/2}+x-\cos\psi}{\sin\psi}\right)$$

$$\int\left(\frac{x^2}{Q^{3/2}}-\cos\psi\frac{x^3}{Q^{3/2}}\right)dx$$

$$=-\frac{(1-2\cos^2\psi)x+\cos\psi+\cos\psi\,(1-\cos^2\psi)x^2-\cos^2\psi\,(5-6\cos^2\psi)x+2\cos\psi-3\cos^3\psi}{(1-\cos^2\psi)Q^{\frac{1}{2}}}$$

$$+(1-3\cos^2\psi)\ln\left(\frac{Q^{1/2}+x-\cos\psi}{\sin\psi}\right)$$

$$=-\frac{(1-2\cos^2\psi)x+\cos\psi\,(1-\cos^2\psi)x^2-\cos^2\psi\,(5-6\cos^2\psi)x+3\cos\psi-3\cos^3\psi}{(1-\cos^2\psi)Q^{\frac{1}{2}}}$$

$$+(1-3\cos^2\psi)\ln\left(\frac{Q^{1/2}+x-\cos\psi}{\sin\psi}\right)$$

$$=-\frac{(1-\cos^2\psi)x-x\cos^2\psi+\cos\psi\,(1-\cos^2\psi)x^2-\cos^2\psi\,(5-6\cos^2\psi)x+3\cos\psi-3\cos^3\psi}{(1-\cos^2\psi)Q^{\frac{1}{2}}}$$

$$+(1-3\cos^2\psi)\ln\left(\frac{Q^{1/2}+x-\cos\psi}{\sin\psi}\right)$$

$$=-\frac{(1-\cos^2\psi)x+\cos\psi\,(1-\cos^2\psi)x^2-x\cos^2\psi-\cos^2\psi\,(5-6\cos^2\psi)x+3\cos\psi-3\cos^3\psi}{(1-\cos^2\psi)Q^{\frac{1}{2}}}$$

$$+(1-3\cos^2\psi)\ln\left(\frac{Q^{1/2}+x-\cos\psi}{\sin\psi}\right)$$

$$=-\frac{(1-\cos^2\psi)x+\cos\psi\,(1-\cos^2\psi)x^2-\cos^2\psi\,(6-6\cos^2\psi)x+3\cos\psi-3\cos^3\psi}{(1-\cos^2\psi)Q^{\frac{1}{2}}}$$

$$+(1-3\cos^2\psi)\ln\left(\frac{Q^{\frac{1}{2}}+x-\cos\psi}{\sin\psi}\right)$$

$$= -\frac{(1-\cos^2\psi)x + \cos\psi\,(1-\cos^2\psi)x^2 - 6\cos^2\psi\,(1-\cos^2\psi)x + 3\cos\psi\,(1-\cos^2\psi)}{(1-\cos^2\psi)Q^{\frac{1}{2}}}$$

$$+ (1-3\cos^2\psi)\ln\left(\frac{Q^{\frac{1}{2}} + x - \cos\psi}{\sin\psi}\right)$$

$$= -\frac{x + x^2\cos\psi - 6\,x\cos^2\psi + 3\cos\psi}{Q^{\frac{1}{2}}}$$

$$+ (1-3\cos^2\psi)\ln\left(\frac{Q^{1/2} + x - \cos\psi}{\sin\psi}\right)$$

$$= -\frac{x - 2x^2\cos\psi + 3x^2\cos\psi - 6\,x\cos^2\psi + 3\cos\psi}{Q^{\frac{1}{2}}}$$

$$+ (1-3\cos^2\psi)\ln\left(\frac{Q^{1/2} + x - \cos\psi}{\sin\psi}\right)$$

$$= -\frac{x - 2x^2\cos\psi + 3\cos\psi\,(x^2 - 2x\cos\psi + 1)}{Q^{\frac{1}{2}}}$$

$$+ (1-3\cos^2\psi)\ln\left(\frac{Q^{1/2} + x - \cos\psi}{\sin\psi}\right)$$

$$= -\frac{x - 2x^2\cos\psi + 3\cos\psi\,Q}{Q^{\frac{1}{2}}} + (1-3\cos^2\psi)\ln\left(\frac{Q^{1/2} + x - \cos\psi}{\sin\psi}\right)$$

$$= -\frac{x - 2x^2\cos\psi}{Q^{\frac{1}{2}}} - 3\cos\psi\,Q^{\frac{1}{2}} + (1-3\cos^2\psi)\ln\left(\frac{Q^{1/2} + x - \cos\psi}{\sin\psi}\right)$$

$$= -\frac{-x^3 + x^3 + x - 2x^2\cos\psi}{Q^{\frac{1}{2}}} - 3\cos\psi\,Q^{\frac{1}{2}}$$

$$+ (1-3\cos^2\psi)\ln\left(\frac{Q^{1/2} + x - \cos\psi}{\sin\psi}\right)$$

$$= -\frac{-x^3 + x(x^2 + 1 - 2x\cos\psi)}{Q^{\frac{1}{2}}} - 3\cos\psi\,Q^{\frac{1}{2}}$$

$$+ (1-3\cos^2\psi)\ln\left(\frac{Q^{1/2} + x - \cos\psi}{\sin\psi}\right)$$

$$= -\frac{-x^3 + xQ}{Q^{\frac{1}{2}}} - 3\cos\psi\,Q^{\frac{1}{2}} + (1-3\cos^2\psi)\ln\left(\frac{Q^{1/2} + x - \cos\psi}{\sin\psi}\right)$$

$$= \frac{x^3}{Q^{\frac{1}{2}}} - xQ^{\frac{1}{2}} - 3\cos\psi\,Q^{\frac{1}{2}} + (1-3\cos^2\psi)\ln\left(\frac{Q^{1/2} + x - \cos\psi}{\sin\psi}\right)$$

$$= \frac{x^3}{Q^{\frac{1}{2}}} - (x + 3\cos\psi)Q^{\frac{1}{2}} + (1 - 3\cos^2\psi)\ln\left(\frac{Q^{1/2} + x - \cos\psi}{\sin\psi}\right)$$

$$= \frac{x^3}{(1 + x^2 - 2x\cos\psi)^{\frac{1}{2}}} - (x + 3\cos\psi)(1 + x^2 - 2x\cos\psi)^{\frac{1}{2}}$$
$$+ (1 - 3\cos^2\psi)\ln\left[\frac{(1 + x^2 - 2x\cos\psi)^{1/2} + x - \cos\psi}{\sin\psi}\right]$$

Eq. (6.90)—(6.92)

$$f(x) = \frac{x^3}{(1 + x^2 - 2x\cos\psi)^{\frac{1}{2}}} - (x + 3\cos\psi)(1 + x^2 - 2x\cos\psi)^{\frac{1}{2}}$$
$$+ (1 - 3\cos^2\psi)\ln\left[(1 + x^2 - 2x\cos\psi)^{\frac{1}{2}} + x - \cos\psi\right]$$

$$f'(x) = \frac{3x^2}{(1 + x^2 - 2x\cos\psi)^{\frac{1}{2}}} - \frac{x^3(x - \cos\psi)}{(1 + x^2 - 2x\cos\psi)^{\frac{3}{2}}} - (1 + x^2 - 2x\cos\psi)^{\frac{1}{2}}$$
$$- \frac{(x + 3\cos\psi)(x - \cos\psi)}{(1 + x^2 - 2x\cos\psi)^{\frac{1}{2}}} + \frac{(1 - 3\cos^2\psi)\left[\frac{x - \cos\psi}{(1 + x^2 - 2x\cos\psi)^{\frac{1}{2}}} + 1\right]}{(1 + x^2 - 2x\cos\psi)^{\frac{1}{2}} + x - \cos\psi}$$
$$= \frac{3x^2}{(1 + x^2 - 2x\cos\psi)^{\frac{1}{2}}} - \frac{x^3(x - \cos\psi)}{(1 + x^2 - 2x\cos\psi)^{\frac{3}{2}}} - (1 + x^2 - 2x\cos\psi)^{\frac{1}{2}}$$
$$- \frac{(x + 3\cos\psi)(x - \cos\psi)}{(1 + x^2 - 2x\cos\psi)^{\frac{1}{2}}} + \frac{(1 - 3\cos^2\psi)}{(1 + x^2 - 2x\cos\psi)^{\frac{1}{2}}}$$
$$= \frac{3x^2}{(1 + x^2 - 2x\cos\psi)^{\frac{1}{2}}} + \frac{(1 - 3\cos^2\psi)}{(1 + x^2 - 2x\cos\psi)^{\frac{1}{2}}} - \frac{(x + 3\cos\psi)(x - \cos\psi)}{(1 + x^2 - 2x\cos\psi)^{\frac{1}{2}}}$$
$$- \frac{x^3(x - \cos\psi)}{(1 + x^2 - 2x\cos\psi)^{\frac{3}{2}}} - (1 + x^2 - 2x\cos\psi)^{\frac{1}{2}}$$
$$= \frac{3x^2 + 1 - 3\cos^2\psi - x^2 + x\cos\psi - 3x\cos\psi + 3\cos^2\psi}{(1 + x^2 - 2x\cos\psi)^{\frac{1}{2}}}$$
$$- \frac{x^3(x - \cos\psi)}{(1 + x^2 - 2x\cos\psi)^{\frac{3}{2}}} - (1 + x^2 - 2x\cos\psi)^{\frac{1}{2}}$$

$$f'(x) = \frac{2x^2 + 1 - 2x\cos\psi}{(1 + x^2 - 2x\cos\psi)^{\frac{1}{2}}} - \frac{x^3(x - \cos\psi)}{(1 + x^2 - 2x\cos\psi)^{\frac{3}{2}}} - (1 + x^2 - 2x\cos\psi)^{\frac{1}{2}}$$

$$= \frac{x^2}{(1 + x^2 - 2x\cos\psi)^{\frac{1}{2}}} + \frac{x^2 + 1 - 2x\cos\psi}{(1 + x^2 - 2x\cos\psi)^{\frac{1}{2}}} - \frac{x^3(x - \cos\psi)}{(1 + x^2 - 2x\cos\psi)^{\frac{3}{2}}}$$

$$- (1 + x^2 - 2x\cos\psi)^{\frac{1}{2}}$$

$$= \frac{x^2}{(1 + x^2 - 2x\cos\psi)^{\frac{1}{2}}} + (1 + x^2 - 2x\cos\psi)^{\frac{1}{2}} - \frac{x^3(x - \cos\psi)}{(1 + x^2 - 2x\cos\psi)^{\frac{3}{2}}}$$

$$- (1 + x^2 - 2x\cos\psi)^{\frac{1}{2}} = \frac{x^2}{(1 + x^2 - 2x\cos\psi)^{\frac{1}{2}}} - \frac{x^3(x - \cos\psi)}{(1 + x^2 - 2x\cos\psi)^{\frac{3}{2}}}$$

$$f''^{(x)} = \frac{2x}{(1 + x^2 - 2x\cos\psi)^{\frac{1}{2}}} - \frac{x^2(x - \cos\psi)}{(1 + x^2 - 2x\cos\psi)^{\frac{3}{2}}} - \frac{3x^2(x - \cos\psi) + x^3}{(1 + x^2 - 2x\cos\psi)^{\frac{3}{2}}}$$

$$+ \frac{3x^3(x - \cos\psi)^2}{(1 + x^2 - 2x\cos\psi)^{\frac{5}{2}}}$$

$$= \frac{2x}{(1 + x^2 - 2x\cos\psi)^{\frac{1}{2}}} - \frac{4x^2(x - \cos\psi) + x^3}{(1 + x^2 - 2x\cos\psi)^{\frac{3}{2}}} + \frac{3x^3(x - \cos\psi)^2}{(1 + x^2 - 2x\cos\psi)^{\frac{5}{2}}}$$

$$f'''^{(x)} = \frac{2}{(1 + x^2 - 2x\cos\psi)^{\frac{1}{2}}} - \frac{2x(x - \cos\psi)}{(1 + x^2 - 2x\cos\psi)^{\frac{3}{2}}} - \frac{8x(x - \cos\psi) + 4x^2 + 3x^2}{(1 + x^2 - 2x\cos\psi)^{\frac{3}{2}}}$$

$$+ \frac{12x^2(x - \cos\psi)^2 + 3x^3(x - \cos\psi)}{(1 + x^2 - 2x\cos\psi)^{\frac{5}{2}}} + \frac{9x^2(x - \cos\psi)^2 + 6x^3(x - \cos\psi)}{(1 + x^2 - 2x\cos\psi)^{\frac{5}{2}}}$$

$$- \frac{15x^3(x - \cos\psi)^3}{(1 + x^2 - 2x\cos\psi)^{\frac{7}{2}}}$$

$$= \frac{2}{(1 + x^2 - 2x\cos\psi)^{\frac{1}{2}}} - \frac{10x(x - \cos\psi) + 7x^2}{(1 + x^2 - 2x\cos\psi)^{\frac{3}{2}}}$$

$$+ \frac{21x^2(x - \cos\psi)^2 + 9x^3(x - \cos\psi)}{(1 + x^2 - 2x\cos\psi)^{\frac{5}{2}}} - \frac{15x^3(x - \cos\psi)^3}{(1 + x^2 - 2x\cos\psi)^{\frac{7}{2}}}$$

Eq. (6.90)—(6.92)

alternative using result (6.65)—(6.67)

$$f(x) = \frac{x^3}{(1 + x^2 - 2x\cos\psi)^{\frac{1}{2}}}$$

$$f'^{(x)} = \frac{3x^2}{(1 + x^2 - 2x\cos\psi)^{\frac{1}{2}}} - \frac{x^3(x - \cos\psi)}{(1 + x^2 - 2x\cos\psi)^{\frac{3}{2}}}$$

$$\begin{aligned}
f''^{(x)} &= \frac{6x}{(1 + x^2 - 2x\cos\psi)^{\frac{1}{2}}} - \frac{3x^2(x - \cos\psi)}{(1 + x^2 - 2x\cos\psi)^{\frac{3}{2}}} - \frac{3x^2(x - \cos\psi) + x^3}{(1 + x^2 - 2x\cos\psi)^{\frac{3}{2}}} \\
&\quad + \frac{3x^3(x - \cos\psi)^2}{(1 + x^2 - 2x\cos\psi)^{\frac{5}{2}}} \\
&= \frac{6x}{(1 + x^2 - 2x\cos\psi)^{\frac{1}{2}}} - \frac{6x^2(x - \cos\psi) + x^3}{(1 + x^2 - 2x\cos\psi)^{\frac{3}{2}}} + \frac{3x^3(x - \cos\psi)^2}{(1 + x^2 - 2x\cos\psi)^{\frac{5}{2}}}
\end{aligned}$$

$$\begin{aligned}
f'''^{(x)} &= \frac{6}{(1 + x^2 - 2x\cos\psi)^{\frac{1}{2}}} - \frac{6x(x - \cos\psi)}{(1 + x^2 - 2x\cos\psi)^{\frac{3}{2}}} - \frac{12x(x - \cos\psi) + 6x^2 + 3x^2}{(1 + x^2 - 2x\cos\psi)^{\frac{3}{2}}} \\
&\quad + \frac{18x^2(x - \cos\psi)^2 + 3x^3(x - \cos\psi)}{(1 + x^2 - 2x\cos\psi)^{\frac{5}{2}}} + \frac{9x^2(x - \cos\psi)^2 + 6x^3(x - \cos\psi)}{(1 + x^2 - 2x\cos\psi)^{\frac{5}{2}}} \\
&\quad - \frac{15x^3(x - \cos\psi)^3}{(1 + x^2 - 2x\cos\psi)^{\frac{7}{2}}} \\
&= \frac{6}{(1 + x^2 - 2x\cos\psi)^{\frac{1}{2}}} - \frac{18x(x - \cos\psi) + 9x^2}{(1 + x^2 - 2x\cos\psi)^{\frac{3}{2}}} \\
&\quad + \frac{27x^2(x - \cos\psi)^2 + 9x^3(x - \cos\psi)}{(1 + x^2 - 2x\cos\psi)^{\frac{5}{2}}} - \frac{15x^3(x - \cos\psi)^3}{(1 + x^2 - 2x\cos\psi)^{\frac{7}{2}}}
\end{aligned}$$

XXXXXXXXXXXXXXXXXXXXXXX

$$\frac{3x^2}{(1+x^2-2x\cos\psi)^{\frac{1}{2}}}-\frac{x^3(x-\cos\psi)}{(1+x^2-2x\cos\psi)^{\frac{3}{2}}}-\frac{2x^2}{(1+x^2-2x\,cos\,\psi)^{\frac{1}{2}}}$$
$$=\frac{x^2}{(1+x^2-2x\cos\psi)^{\frac{1}{2}}}-\frac{x^3(x-\cos\psi)}{(1+x^2-2x\cos\psi)^{\frac{3}{2}}}$$

$$\frac{6x}{(1+x^2-2x\cos\psi)^{\frac{1}{2}}}-\frac{6x^2(x-\cos\psi)+x^3}{(1+x^2-2x\cos\psi)^{\frac{3}{2}}}+\frac{3x^3(x-\cos\psi)^2}{(1+x^2-2x\cos\psi)^{\frac{5}{2}}}$$
$$-\frac{4x+2x^3-6x^2\,cos\,\psi}{(1+x^2-2x\,cos\,\psi)^{\frac{3}{2}}}$$
$$=\frac{6x}{(1+x^2-2x\cos\psi)^{\frac{1}{2}}}-\frac{6x^2(x-\cos\psi)+x^3}{(1+x^2-2x\cos\psi)^{\frac{3}{2}}}-\frac{4x+2x^3-6x^2\,cos\,\psi}{(1+x^2-2x\,cos\,\psi)^{\frac{3}{2}}}$$
$$+\frac{3x^3(x-\cos\psi)^2}{(1+x^2-2x\cos\psi)^{\frac{5}{2}}}$$
$$=\frac{6x}{(1+x^2-2x\cos\psi)^{\frac{1}{2}}}-\frac{6x^3-6x^2\cos\psi+x^3+4x+2x^3-6x^2\,cos\,\psi}{(1+x^2-2x\cos\psi)^{\frac{3}{2}}}$$
$$+\frac{3x^3(x-\cos\psi)^2}{(1+x^2-2x\cos\psi)^{\frac{5}{2}}}$$
$$=\frac{6x}{(1+x^2-2x\cos\psi)^{\frac{1}{2}}}-\frac{9x^3-12x^2\cos\psi+4x}{(1+x^2-2x\cos\psi)^{\frac{3}{2}}}+\frac{3x^3(1-\cos\psi)^2}{(1+x^2-2x\cos\psi)^{\frac{5}{2}}}$$
$$=\frac{6x}{(1+x^2-2x\cos\psi)^{\frac{1}{2}}}-\frac{4x+4x^3-8x^2\cos\psi+5x^3-4x^2\cos\psi}{(1+x^2-2x\cos\psi)^{\frac{3}{2}}}$$
$$+\frac{3x^3(x-\cos\psi)^2}{(1+x^2-2x\cos\psi)^{\frac{5}{2}}}$$
$$=\frac{6x}{(1+x^2-2x\cos\psi)^{\frac{1}{2}}}-\frac{4x(1+x^2-2x\cos\psi)+5x^3-4x^2\cos\psi}{(1+x^2-2x\cos\psi)^{\frac{3}{2}}}$$
$$+\frac{3x^3(x-\cos\psi)^2}{(1+x^2-2x\cos\psi)^{\frac{5}{2}}}$$
$$=\frac{6x}{(1+x^2-2x\cos\psi)^{\frac{1}{2}}}-\frac{4x}{(1+x^2-2x\cos\psi)^{\frac{1}{2}}}-\frac{5x^3-4x^2\cos\psi}{(1+x^2-2x\cos\psi)^{\frac{3}{2}}}$$
$$+\frac{3x^3(x-\cos\psi)^2}{(1+x^2-2x\cos\psi)^{\frac{5}{2}}}$$

$$= \frac{2x}{(1+x^2-2x\cos\psi)^{\frac{1}{2}}} - \frac{4x^3-4x^2\cos\psi+x^3}{(1+x^2-2x\cos\psi)^{\frac{3}{2}}} + \frac{3x^3(x-\cos\psi)^2}{(1+x^2-2x\cos\psi)^{\frac{5}{2}}}$$

$$= \frac{2x}{(1+x^2-2x\cos\psi)^{\frac{1}{2}}} - \frac{4x^2(x-4\cos\psi)+x^3}{(1+x^2-2x\cos\psi)^{\frac{3}{2}}} + \frac{3x^3(x-\cos\psi)^2}{(1+x^2-2x\cos\psi)^{\frac{5}{2}}}$$

$$\frac{6}{(1+x^2-2x\cos\psi)^{\frac{1}{2}}} - \frac{18x(x-\cos\psi)+9x^2}{(1+x^2-2x\cos\psi)^{\frac{3}{2}}} + \frac{27x^2(x-\cos\psi)^2+9x^3(x-\cos\psi)}{(1+x^2-2x\cos\psi)^{\frac{5}{2}}}$$

$$- \frac{15x^3(x-\cos\psi)^3}{(1+x^2-2x\cos\psi)^{\frac{7}{2}}}$$

$$- \left[\frac{3}{2(1+x^2-2x\,cos\,\psi)^{\frac{1}{2}}} + \frac{5-9x^2}{2(1+x^2-2x\,cos\,\psi)^{\frac{3}{2}}} - \frac{3x(1-x^2)(x-\cos\psi)}{(1+x^2-2x\,cos\,\psi)^{\frac{5}{2}}}\right]$$

$$= \frac{6}{(1+x^2-2x\cos\psi)^{\frac{1}{2}}} - \frac{3}{2(1+x^2-2x\,cos\,\psi)^{\frac{1}{2}}} - \frac{18x(x-\cos\psi)+9x^2}{(1+x^2-2x\cos\psi)^{\frac{3}{2}}}$$

$$- \frac{5-9x^2}{2(1+x^2-2x\,cos\,\psi)^{\frac{3}{2}}} + \frac{27x^2(x-\cos\psi)^2+9x^3(x-\cos\psi)}{(1+x^2-2x\cos\psi)^{\frac{5}{2}}}$$

$$+ \frac{3x(1-x^2)(x-\cos\psi)}{(1+x^2-2x\,cos\,\psi)^{\frac{5}{2}}} - \frac{15x^3(x-\cos\psi)^3}{(1+x^2-2x\cos\psi)^{\frac{7}{2}}}$$

$$= \frac{9}{2(1+x^2-2x\cos\psi)^{\frac{1}{2}}} - \frac{36x(x-\cos\psi)+5+9x^2}{2(1+x^2-2x\cos\psi)^{\frac{3}{2}}}$$

$$+ \frac{27x^2(x-\cos\psi)^2+[9x^3+3x(1-x^2)](x-\cos\psi)}{(1+x^2-2x\cos\psi)^{\frac{5}{2}}} - \frac{15x^3(x-\cos\psi)^3}{(1+x^2-2x\cos\psi)^{\frac{7}{2}}}$$

$$= \frac{9}{2(1+x^2-2x\cos\psi)^{\frac{1}{2}}} - \frac{36x^2-36x\cos\psi+5+9x^2}{2(1+x^2-2x\cos\psi)^{\frac{3}{2}}}$$

$$+ \frac{27x^2(x-\cos\psi)^2+[9x^3+3x-3x^3](x-\cos\psi)}{(1+x^2-2x\cos\psi)^{\frac{5}{2}}} - \frac{15x^3(x-\cos\psi)^3}{(1+x^2-2x\cos\psi)^{\frac{7}{2}}}$$

$$= \frac{9}{2(1+x^2-2x\cos\psi)^{\frac{1}{2}}} - \frac{5+5x^2-10\cos\psi+40x^2-26x\cos\psi}{2(1+x^2-2x\cos\psi)^{\frac{3}{2}}}$$

$$+ \frac{27x^2(x-\cos\psi)^2+[6x^3+3x](x-\cos\psi)}{(1+x^2-2x\cos\psi)^{\frac{5}{2}}} - \frac{15x^3(x-\cos\psi)^3}{(1+x^2-2x\cos\psi)^{\frac{7}{2}}}$$

$$
\begin{aligned}
&= \frac{9}{2(1+x^2-2x\cos\psi)^{\frac{1}{2}}} - \frac{5}{2(1+x^2-2x\cos\psi)^{\frac{1}{2}}} - \frac{40x^2-26x\cos\psi}{2(1+x^2-2x\cos\psi)^{\frac{3}{2}}} \\
&\quad + \frac{27x^2(x-\cos\psi)^2 + [6x^3+3x](x-\cos\psi)}{(1+x^2-2x\cos\psi)^{\frac{5}{2}}} - \frac{15x^3(x-\cos\psi)^3}{(1+x^2-2x\cos\psi)^{\frac{7}{2}}} \\
&= \frac{2}{(1+x^2-2x\cos\psi)^{\frac{1}{2}}} - \frac{40x^2-26x\cos\psi}{2(1+x^2-2x\cos\psi)^{\frac{3}{2}}} \\
&\quad + \frac{27x^2(x-\cos\psi)^2 + [6x^3+3x](x-\cos\psi)}{(1+x^2-2x\cos\psi)^{\frac{5}{2}}} - \frac{15x^3(x-\cos\psi)^3}{(1+x^2-2x\cos\psi)^{\frac{7}{2}}} \\
&= \frac{2}{(1+x^2-2x\cos\psi)^{\frac{1}{2}}} - \frac{20x^2-13x\cos\psi}{(1+x^2-2x\cos\psi)^{\frac{3}{2}}} \\
&\quad + \frac{27x^2(x-\cos\psi)^2 + [6x^3+3x](x-\cos\psi)}{(1+x^2-2x\cos\psi)^{\frac{5}{2}}} - \frac{15x^3(x-\cos\psi)^3}{(1+x^2-2x\cos\psi)^{\frac{7}{2}}} \\
&= \frac{2}{(1+x^2-2x\cos\psi)^{\frac{1}{2}}} - \frac{10x(x-\cos\psi)+7x^2+3x^2-3x\cos\psi}{(1+x^2-2x\cos\psi)^{\frac{3}{2}}} \\
&\quad + \frac{27x^2(x-\cos\psi)^2 + [6x^3+3x](x-\cos\psi)}{(1+x^2-2x\cos\psi)^{\frac{5}{2}}} - \frac{15x^3(x-\cos\psi)^3}{(1+x^2-2x\cos\psi)^{\frac{7}{2}}} \\
&= \frac{2}{(1+x^2-2x\cos\psi)^{\frac{1}{2}}} - \frac{10x(x-\cos\psi)+7x^2}{(1+x^2-2x\cos\psi)^{\frac{3}{2}}} - \frac{3x^2-3x\cos\psi}{(1+x^2-2x\cos\psi)^{\frac{3}{2}}} \\
&\quad + \frac{21x^2(x-\cos\psi)^2 + 9x^3(x-\cos\psi)}{(1+x^2-2x\cos\psi)^{\frac{5}{2}}} \\
&\quad + \frac{6x^2(x-\cos\psi)^2 + [3x-3x^3](x-\cos\psi)}{(1+x^2-2x\cos\psi)^{\frac{5}{2}}} - \frac{15x^3(x-\cos\psi)^3}{(1+x^2-2x\cos\psi)^{\frac{7}{2}}} \\
&= \frac{2}{(1+x^2-2x\cos\psi)^{\frac{1}{2}}} - \frac{10x(x-\cos\psi)+7x^2}{(1+x^2-2x\cos\psi)^{\frac{3}{2}}} \\
&\quad + \frac{21x^2(x-\cos\psi)^2 + 9x^3(x-\cos\psi)}{(1+x^2-2x\cos\psi)^{\frac{5}{2}}} - \frac{15x^3(x-\cos\psi)^3}{(1+x^2-2x\cos\psi)^{\frac{7}{2}}} \\
&\quad - \frac{3x^2-3x\cos\psi}{(1+x^2-2x\cos\psi)^{\frac{3}{2}}} + \frac{6x^2(x-\cos\psi)^2 + [3x-3x^3](x-\cos\psi)}{(1+x^2-2x\cos\psi)^{\frac{5}{2}}} \\
&= \frac{2}{(1+x^2-2x\cos\psi)^{\frac{1}{2}}} - \frac{10x(x-\cos\psi)+7x^2}{(1+x^2-2x\cos\psi)^{\frac{3}{2}}} \\
&\quad + \frac{21x^2(x-\cos\psi)^2 + 9x^3(x-\cos\psi)}{(1+x^2-2x\cos\psi)^{\frac{5}{2}}} - \frac{15x^3(x-\cos\psi)^3}{(1+x^2-2x\cos\psi)^{\frac{7}{2}}}
\end{aligned}
$$

$$-\frac{3x^2 - 3x\cos\psi}{(1 + x^2 - 2x\cos\psi)^{\frac{3}{2}}} + \frac{[6x^3 - 6x^2\cos\psi + 3x - 3x^3](x - \cos\psi)}{(1 + x^2 - 2x\cos\psi)^{\frac{5}{2}}}$$

$$= \frac{2}{(1 + x^2 - 2x\cos\psi)^{\frac{1}{2}}} - \frac{10x(x - \cos\psi) + 7x^2}{(1 + x^2 - 2x\cos\psi)^{\frac{3}{2}}}$$

$$+ \frac{21x^2(x - \cos\psi)^2 + 9x^3(x - \cos\psi)}{(1 + x^2 - 2x\cos\psi)^{\frac{5}{2}}} - \frac{15x^3(x - \cos\psi)^3}{(1 + x^2 - 2x\cos\psi)^{\frac{7}{2}}}$$

$$-\frac{3x^2 - 3x\cos\psi}{(1 + x^2 - 2x\cos\psi)^{\frac{3}{2}}} + \frac{[3x^3 - 6x^2\cos\psi + 3x](x - \cos\psi)}{(1 + x^2 - 2x\cos\psi)^{\frac{5}{2}}}$$

$$= \frac{2}{(1 + x^2 - 2x\cos\psi)^{\frac{1}{2}}} - \frac{10x(x - \cos\psi) + 7x^2}{(1 + x^2 - 2x\cos\psi)^{\frac{3}{2}}}$$

$$+ \frac{21x^2(x - \cos\psi)^2 + 9x^3(x - \cos\psi)}{(1 + x^2 - 2x\cos\psi)^{\frac{5}{2}}} - \frac{15x^3(x - \cos\psi)^3}{(1 + x^2 - 2x\cos\psi)^{\frac{7}{2}}}$$

$$-\frac{3x^2 - 3x\cos\psi}{(1 + x^2 - 2x\cos\psi)^{\frac{3}{2}}} + \frac{3x(x - \cos\psi)}{(1 + x^2 - 2x\cos\psi)^{\frac{3}{2}}}$$

$$= \frac{2}{(1 + x^2 - 2x\cos\psi)^{\frac{1}{2}}} - \frac{10x(x - \cos\psi) + 7x^2}{(1 + x^2 - 2x\cos\psi)^{\frac{3}{2}}}$$

$$+ \frac{21x^2(x - \cos\psi)^2 + 9x^3(x - \cos\psi)}{(1 + x^2 - 2x\cos\psi)^{\frac{5}{2}}} - \frac{15x^3(x - \cos\psi)^3}{(1 + x^2 - 2x\cos\psi)^{\frac{7}{2}}}$$

Lengthy Derivations in Sect. 6.3.2

Basic Integrals

2.26 Forms containing $\sqrt{a + bx + cx^2}$ and integral powers of x

Notation: $R = a + bx + cx^2, \quad \Delta = 4ac - b^2$

2.261[11] For $n = -1$

$$\int \frac{dx}{\sqrt{R}} = \frac{1}{\sqrt{c}} \ln\left(\frac{2\sqrt{cR} + 2cx + b}{\sqrt{\Delta}}\right) \qquad [c > 0]$$

$$\int x\sqrt{R}\,dx = \frac{\sqrt{R^3}}{3c} - \frac{(2cx+b)b}{8c^2}\sqrt{R} - \frac{b\Delta}{16c^2}\int\frac{dx}{\sqrt{R}} \qquad (\text{see } \mathbf{2.261})$$

2.264

1. $$\int \frac{dx}{\sqrt{R}} \qquad (\text{see } \mathbf{2.261})$$

2. $$\int \frac{x\,dx}{\sqrt{R}} = \frac{\sqrt{R}}{c} - \frac{b}{2c}\int\frac{dx}{\sqrt{R}} \qquad (\text{see } \mathbf{2.261})$$

3. $$\int \frac{x^2\,dx}{\sqrt{R}} = \left(\frac{x}{2c} - \frac{3b}{4c^2}\right)\sqrt{R} + \left(\frac{3b^2}{8c^2} - \frac{a}{2c}\right)\int\frac{dx}{\sqrt{R}} \qquad (\text{see } \mathbf{2.261})$$

4. $$\int \frac{x^3\,dx}{\sqrt{R}} = \left(\frac{x^2}{3c} - \frac{5bx}{12c^2} + \frac{5b^2}{8c^3} - \frac{2a}{3c^2}\right)\sqrt{R} - \left(\frac{5b^3}{16c^3} - \frac{3ab}{4c^2}\right)\int\frac{dx}{\sqrt{R}}$$

(see **2.261**)

$$a = 1, \qquad b = -2\cos\Theta, \qquad c = 1, \qquad \Delta = 4(1 - \cos^2\Theta) = 4\sin^2\Theta$$

$$\int x\sqrt{R}\,dx = \frac{1}{3}R^{\frac{3}{2}} - \frac{(2x - 2\cos\Theta)(-2\cos\Theta)}{8}R^{\frac{1}{2}} - \frac{(-2\cos\Theta)(4\sin^2\Theta)}{16}\ln\left(\frac{2R^{\frac{1}{2}} + 2x - 2\cos\Theta}{2\sin\Theta}\right) + C$$

$$\int x\sqrt{R}\,dx = \frac{1}{3}R^{\frac{3}{2}} + \frac{(x - \cos\Theta)\cos\Theta}{2}R^{\frac{1}{2}} + \frac{\sin^2\Theta\cos\Theta}{2}\ln\left(\frac{R^{\frac{1}{2}} + x - \cos\Theta}{\sin\theta}\right) + C$$

$\sin\theta$ in the denominator within the ln function contributes to a constant, and can be dropped.

$$\int x(1+x^2-2x\cos\Theta)^{\frac{1}{2}}\,dx$$
$$=\frac{1}{3}(1+x^2-2x\cos\Theta)^{\frac{3}{2}}+\frac{1}{2}(x-\cos\Theta)\cos\Theta\,(1+x^2-2x\cos\Theta)^{\frac{1}{2}}$$
$$+\frac{1}{2}\sin^2\Theta\cos\Theta\ln\left[(1+x^2-2x\cos\Theta)^{\frac{1}{2}}+x-\cos\Theta\right]+C$$

$$\int\frac{x^3dx}{\sqrt{R}}=\left(\frac{x^2}{3}+\frac{5x\cos\Theta}{6}+\frac{5\cos^2\Theta}{2}-\frac{2}{3}\right)\sqrt{R}+\left(\frac{5\cos^3\Theta}{2}-\frac{3\cos\Theta}{2}\right)\ln\left(\frac{\sqrt{R}+x-\cos\Theta}{\sin\Theta}\right)$$
$$+C$$

$$\int\frac{x^3dx}{\sqrt{R}}=\left(\frac{x^2}{3}+\frac{5x\cos\Theta}{6}+\frac{5\cos^2\Theta}{2}-\frac{2}{3}\right)(1+x^2-2x\cos\Theta)^{\frac{1}{2}}$$
$$+\left(\frac{5\cos^3\Theta}{2}-\frac{3\cos\Theta}{2}\right)\ln\left[(1+x^2-2x\cos\Theta)^{\frac{1}{2}}+x-\cos\Theta\right]+C$$

$$\int\frac{x^2dx}{\sqrt{R}}=\left(\frac{x}{2}+\frac{3\cos\Theta}{2}\right)\sqrt{R}+\left(\frac{3\cos^2\Theta}{2}-\frac{1}{2}\right)\ln\left(\frac{\sqrt{R}+x-\cos\Theta}{\sin\Theta}\right)$$

$$\int\frac{x^2dx}{\sqrt{R}}=\left(\frac{x}{2}+\frac{3\cos\Theta}{2}\right)(1+x^2-2x\cos\Theta)^{\frac{1}{2}}$$
$$+\left(\frac{3\cos^2\Theta}{2}-\frac{1}{2}\right)\ln\left[(1+x^2-2x\cos\Theta)^{\frac{1}{2}}+x-\cos\Theta\right]+C$$

Eq. (6.117)

$$\int\frac{x^3dx}{\sqrt{R}}-\cos\Theta\int\frac{x^2dx}{\sqrt{R}}$$
$$=\left(\frac{x^2}{3}+\frac{5x\cos\Theta}{6}+\frac{5\cos^2\Theta}{2}-\frac{2}{3}-\frac{x\cos\Theta}{2}-\frac{3\cos^2\Theta}{2}\right)\sqrt{R}$$
$$+\left(\frac{5\cos^3\Theta}{2}-\frac{3\cos\Theta}{2}-\frac{3\cos^3\Theta}{2}+\frac{\cos\Theta}{2}\right)\ln\left(\sqrt{R}+x-\cos\Theta\right)+C$$

$$\int \frac{x^3 dx}{\sqrt{R}} - \cos\Theta \int \frac{x^2 dx}{\sqrt{R}} = \left(\frac{x^2}{3} + \frac{x\cos\Theta}{3} + \cos^2\Theta - \frac{2}{3}\right)\sqrt{x^2 - 2x\cos\Theta + 1} + (\cos^3\Theta - \cos\Theta)\ln\left(\sqrt{x^2 - 2x\cos\Theta + 1} + x - \cos\Theta\right) + C$$

$$\int \frac{x^3 dx}{\sqrt{R}} - \cos\Theta \int \frac{x^2 dx}{\sqrt{R}} = \left(\frac{x^2 + x\cos\Theta + 1}{3} - \sin^2\Theta\right)\sqrt{x^2 - 2x\cos\Theta + 1} - \sin^2\Theta\cos\Theta\ln\left(\sqrt{x^2 - 2x\cos\Theta + 1} + x - \cos\Theta\right) + C$$

$$\int \frac{x^3 dx}{\sqrt{R}} - \cos\Theta \int \frac{x^2 dx}{\sqrt{R}} = \left(\frac{x^2 + x\cos\Theta + 1}{3} - \sin^2\Theta\right)\sqrt{x^2 - 2x\cos\Theta + 1} - \sin^2\Theta\cos\Theta\ln\left(\sqrt{x^2 - 2x\cos\Theta + 1} + x - \cos\Theta\right) + C$$

$$\int \frac{x^3 dx}{\sqrt{R}} - \cos\Theta \int \frac{x^2 dx}{\sqrt{R}} + \int x^2 dx = \frac{1}{3}x^3 + \left(\frac{x^2 + x\cos\Theta + 1}{3} - \sin^2\Theta\right)\sqrt{x^2 - 2x\cos\Theta + 1} - \sin^2\Theta\cos\Theta\ln\left(\sqrt{x^2 - 2x\cos\Theta + 1} + x - \cos\Theta\right) + C$$

$$\int \frac{x^3 dx}{\sqrt{R}} - \cos\Theta \int \frac{x^2 dx}{\sqrt{R}} + \int x^2 dx = \frac{1}{3}\Big[x^3 + (x^2 + x\cos\Theta + 1 - 3\sin^2\Theta)\sqrt{x^2 - 2x\cos\Theta + 1} - 3\sin^2\Theta\cos\Theta\ln\left(\sqrt{x^2 - 2x\cos\Theta + 1} + x - \cos\Theta\right)\Big] + C$$

Eq. (6.124)—(6.126)

$$f(x) = \frac{1}{3}(1 + x^2 - 2x\cos\Theta)^{\frac{3}{2}} + \frac{1}{2}(x - \cos\Theta)\cos\Theta\,(1 + x^2 - 2x\cos\Theta)^{\frac{1}{2}} - \frac{1}{2}x^2 + \frac{1}{3}x^3 + \frac{1}{2}\sin^2\Theta\cos\Theta\ln\left[(1 + x^2 - 2x\cos\Theta)^{\frac{1}{2}} + x - \cos\Theta\right]$$

$$f'(x) = (1 + x^2 - 2x\cos\Theta)^{\frac{1}{2}}(x - \cos\Theta) + \frac{1}{2}\cos\Theta\,(1 + x^2 - 2x\cos\Theta)^{\frac{1}{2}} + \frac{(x - \cos\Theta)^2\cos\Theta}{2(1 + x^2 - 2x\cos\Theta)^{\frac{1}{2}}}$$

$$- x + x^2 + \frac{1}{2}\sin^2\Theta\cos\Theta\,\frac{\dfrac{(x - \cos\Theta)}{(1 + x^2 - 2x\cos\Theta)^{\frac{1}{2}}} + 1}{(1 + x^2 - 2x\cos\Theta)^{\frac{1}{2}} + x - \cos\Theta}$$

$$= (1 + x^2 - 2x\cos\Theta)^{\frac{1}{2}}(x - \cos\Theta) + \frac{1}{2}\cos\Theta\,(1 + x^2 - 2x\cos\Theta)^{\frac{1}{2}}$$

$$+ \frac{(x - \cos\Theta)^2\cos\Theta}{2(1 + x^2 - 2x\cos\Theta)^{\frac{1}{2}}} - x + x^2 + \frac{\sin^2\Theta\cos\Theta}{2(1 + x^2 - 2x\cos\Theta)^{\frac{1}{2}}}$$

$$= (1 + x^2 - 2x\cos\Theta)^{\frac{1}{2}}(x - \cos\Theta) + \frac{1}{2}\cos\Theta\,(1 + x^2 - 2x\cos\Theta)^{\frac{1}{2}}$$

$$+ \frac{[(x - \cos\Theta)^2 + \sin^2\Theta]\cos\Theta}{2(1 + x^2 - 2x\cos\Theta)^{\frac{1}{2}}} - x + x^2$$

$$= (1 + x^2 - 2x\cos\Theta)^{\frac{1}{2}}(x - \cos\Theta) + \frac{1}{2}\cos\Theta\,(1 + x^2 - 2x\cos\Theta)^{\frac{1}{2}}$$

$$+ \frac{[x^2 - 2x\cos\Theta + \cos^2\Theta + \sin^2\Theta]\cos\Theta}{2(1 + x^2 - 2x\cos\Theta)^{\frac{1}{2}}} - x + x^2$$

$$= (1 + x^2 - 2x\cos\Theta)^{\frac{1}{2}}(x - \cos\Theta) + \frac{1}{2}\cos\Theta\,(1 + x^2 - 2x\cos\Theta)^{\frac{1}{2}}$$

$$+ \frac{[x^2 - 2x\cos\Theta + 1]\cos\Theta}{2(1 + x^2 - 2x\cos\Theta)^{\frac{1}{2}}} - x + x^2$$

$$= (1 + x^2 - 2x\cos\Theta)^{\frac{1}{2}}(x - \cos\Theta) + \frac{1}{2}\cos\Theta\,(1 + x^2 - 2x\cos\Theta)^{\frac{1}{2}}$$

$$+ \frac{1}{2}\cos\Theta\,(1 + x^2 - 2x\cos\Theta)^{\frac{1}{2}} - x + x^2$$

$$= (1 + x^2 - 2x\cos\Theta)^{\frac{1}{2}}(x - \cos\Theta) + \cos\Theta\,(1 + x^2 - 2x\cos\Theta)^{\frac{1}{2}} - x + x^2$$

$$= x(1 + x^2 - 2x\cos\Theta)^{\frac{1}{2}} - x + x^2$$

$$f''(x) = (1 + x^2 - 2x\cos\Theta)^{\frac{1}{2}} + \frac{x(x - \cos\Theta)}{(1 + x^2 - 2x\cos\Theta)^{\frac{1}{2}}} - 1 + 2x$$

$$f'''(x) = \frac{x - \cos\Theta}{(1 + x^2 - 2x\cos\Theta)^{\frac{1}{2}}} + \frac{(x - \cos\Theta) + x}{(1 + x^2 - 2x\cos\Theta)^{\frac{1}{2}}} - \frac{x(x - \cos\Theta)^2}{(1 + x^2 - 2x\cos\Theta)^{\frac{3}{2}}} + 2$$

$$= \frac{x + 2(x - \cos\Theta)}{(1 + x^2 - 2x\cos\Theta)^{\frac{1}{2}}} - \frac{x(x - \cos\Theta)^2}{(1 + x^2 - 2x\cos\Theta)^{\frac{3}{2}}} + 2$$

Eq. (6.127)—(6.129)

$$
\begin{aligned}
3f(x) = \int \frac{x^3 dx}{\sqrt{R}} - \cos\Theta \int \frac{x^2 dx}{\sqrt{R}} + \int x^2 dx \\
&= (x^2 + x\cos\Theta + 1 - 3\sin^2\Theta)\sqrt{x^2 - 2x\cos\Theta + 1} + x^3 \\
&- 3\sin^2\Theta\cos\Theta \ln\left(\sqrt{x^2 - 2x\cos\Theta + 1} + x - \cos\Theta\right)
\end{aligned}
$$

$$
\begin{aligned}
3f'(x) &= (2x + \cos\Theta)\sqrt{x^2 - 2x\cos\Theta + 1} + \frac{(x^2 + x\cos\Theta + 1 - 3\sin^2\Theta)(x - \cos\Theta)}{\sqrt{x^2 - 2x\cos\Theta + 1}} + 3x^2 \\
&- 3\sin^2\Theta\cos\Theta \frac{\dfrac{x - \cos\Theta}{\sqrt{x^2 - 2x\cos\Theta + 1}} + 1}{\sqrt{x^2 - 2x\cos\Theta + 1} + x - \cos\Theta} \\
&= (2x + \cos\Theta)\sqrt{x^2 - 2x\cos\Theta + 1} + \frac{(x^2 + x\cos\Theta + 1 - 3\sin^2\Theta)(x - \cos\Theta)}{\sqrt{x^2 - 2x\cos\Theta + 1}} + 3x^2 \\
&- \frac{3\sin^2\Theta\cos\Theta}{\sqrt{x^2 - 2x\cos\Theta + 1}} \\
&= (2x + \cos\Theta)\sqrt{x^2 - 2x\cos\Theta + 1} + 3x^2 \\
&+ \frac{(x^2 + x\cos\Theta + 1)(x - \cos\Theta) - 3\sin^2\Theta\,(x - \cos\Theta)}{\sqrt{x^2 - 2x\cos\Theta + 1}} - \frac{3\sin^2\Theta\cos\Theta}{\sqrt{x^2 - 2x\cos\Theta + 1}} \\
&= (2x + \cos\Theta)\sqrt{x^2 - 2x\cos\Theta + 1} + 3x^2 \\
&+ \frac{(x^2 + x\cos\Theta + 1)(x - \cos\Theta) - 3x\sin^2\Theta}{\sqrt{x^2 - 2x\cos\Theta + 1}} \\
&= (2x + \cos\Theta)\sqrt{x^2 - 2x\cos\Theta + 1} + 3x^2 \\
&+ \frac{(x^2 - 2x\cos\Theta + 1)(x - \cos\Theta) + 3x\cos\Theta\,(x - \cos\Theta) - 3x\sin^2\Theta}{\sqrt{x^2 - 2x\cos\Theta + 1}} \\
&= (2x + \cos\Theta)\sqrt{x^2 - 2x\cos\Theta + 1} + 3x^2 + (x - \cos\Theta)\sqrt{x^2 - 2x\cos\Theta + 1} \\
&+ \frac{3x\cos\Theta\,(x - \cos\Theta) - 3x\sin^2\Theta}{\sqrt{x^2 - 2x\cos\Theta + 1}} \\
&= (2x + \cos\Theta)\sqrt{x^2 - 2x\cos\Theta + 1} + (x - \cos\Theta)\sqrt{x^2 - 2x\cos\Theta + 1} + 3x^2 \\
&+ \frac{3x^2\cos\Theta - 3x\cos^2\Theta - 3x\sin^2\Theta}{\sqrt{x^2 - 2x\cos\Theta + 1}} \\
&= 3x\sqrt{x^2 - 2x\cos\Theta + 1} + 3x^2 + \frac{3x^2\cos\Theta - 3x}{\sqrt{x^2 - 2x\cos\Theta + 1}} \\
&= 3x\sqrt{x^2 - 2x\cos\Theta + 1} + 3x^2 + \frac{3x(x\cos\Theta - 1)}{\sqrt{x^2 - 2x\cos\Theta + 1}} \quad (used\ for\ f'') \\
&= 3x^2 + 3x\sqrt{x^2 - 2x\cos\Theta + 1} - \frac{3x(1 - x\cos\Theta)}{\sqrt{x^2 - 2x\cos\Theta + 1}}
\end{aligned}
$$

$$3f''(x) = 3\sqrt{x^2 - 2x\cos\Theta + 1} + \frac{3x(x - \cos\Theta)}{\sqrt{x^2 - 2x\cos\Theta + 1}} + 6x + \frac{6x\cos\Theta - 3}{\sqrt{x^2 - 2x\cos\Theta + 1}}$$
$$- \frac{3x(x\cos\Theta - 1)(x - \cos\Theta)}{\left(\sqrt{x^2 - 2x\cos\Theta + 1}\right)^3}$$
$$= 3\sqrt{x^2 - 2x\cos\Theta + 1} + \frac{3x(x - \cos\Theta)}{\sqrt{x^2 - 2x\cos\Theta + 1}} + 6x + \frac{6x\cos\Theta - 3 - 3x^2}{\sqrt{x^2 - 2x\cos\Theta + 1}}$$
$$+ \frac{3x^2}{\sqrt{x^2 - 2x\cos\Theta + 1}} - \frac{3x(x\cos\Theta - 1)(x - \cos\Theta)}{\left(\sqrt{x^2 - 2x\cos\Theta + 1}\right)^3}$$
$$= 3\sqrt{x^2 - 2x\cos\Theta + 1} + \frac{3x(x - \cos\Theta)}{\sqrt{x^2 - 2x\cos\Theta + 1}} + 6x - 3\sqrt{x^2 - 2x\cos\Theta + 1}$$
$$+ \frac{3x^2}{\sqrt{x^2 - 2x\cos\Theta + 1}} - \frac{3x(x\cos\Theta - 1)(x - \cos\Theta)}{\left(\sqrt{x^2 - 2x\cos\Theta + 1}\right)^3}$$
$$= 6x + \frac{3x(x - \cos\Theta)}{\sqrt{x^2 - 2x\cos\Theta + 1}} + \frac{3x^2}{\sqrt{x^2 - 2x\cos\Theta + 1}} - \frac{3x(x\cos\Theta - 1)(x - \cos\Theta)}{\left(\sqrt{x^2 - 2x\cos\Theta + 1}\right)^3}$$
$$= 6x + \frac{3x(2x - \cos\Theta)}{\sqrt{x^2 - 2x\cos\Theta + 1}} + \frac{3x(1 - x\cos\Theta)(x - \cos\Theta)}{\left(\sqrt{x^2 - 2x\cos\Theta + 1}\right)^3}$$

$$3f'''(x)$$
$$= 6 + \frac{3(2x - \cos\Theta) + 6x}{(x^2 - 2x\cos\Theta + 1)^{1/2}} - \frac{3x(2x - \cos\Theta)(x - \cos\Theta)}{(x^2 - 2x\cos\Theta + 1)^{\frac{3}{2}}}$$
$$+ \frac{3(1 - x\cos\Theta)(x - \cos\Theta) - 3x\cos\Theta\,(x - \cos\Theta) + 3x(1 - x\cos\Theta)}{(x^2 - 2x\cos\Theta + 1)^{\frac{3}{2}}} - \frac{9x(1 - x\cos\Theta)(x - \cos\Theta)^2}{(x^2 - 2x\cos\Theta + 1)^{\frac{5}{2}}}$$
$$= 6 + \frac{3(x - \cos\Theta) + 9x}{(x^2 - 2x\cos\Theta + 1)^{1/2}}$$
$$+ \frac{3(1 - x\cos\Theta)(x - \cos\Theta) - 3x\cos\Theta\,(x - \cos\Theta) + 3x(1 - x\cos\Theta) - 3x(2x - \cos\Theta)(x - \cos\Theta)}{(x^2 - 2x\cos\Theta + 1)^{\frac{3}{2}}}$$
$$- \frac{9x(1 - x\cos\Theta)(x - \cos\Theta)^2}{(x^2 - 2x\cos\Theta + 1)^{\frac{5}{2}}}$$
$$= 6 + \frac{3(x - \cos\Theta) + 9x}{(x^2 - 2x\cos\Theta + 1)^{1/2}}$$
$$+ \frac{3(1 - x\cos\Theta)(x - \cos\Theta) + 3x(1 - x\cos\Theta) - 3x(2x - \cos\Theta)(x - \cos\Theta) - 3x\cos\Theta\,(x - \cos\Theta)}{(x^2 - 2x\cos\Theta + 1)^{\frac{3}{2}}}$$
$$- \frac{9x(1 - x\cos\Theta)(x - \cos\Theta)^2}{(x^2 - 2x\cos\Theta + 1)^{\frac{5}{2}}}$$

$$= 6 + \frac{3(x-\cos\Theta)+9x}{(x^2-2x\cos\Theta+1)^{1/2}}$$
$$+ \frac{3(1-x\cos\Theta)(x-\cos\Theta)+3x(1-x\cos\Theta)-3x(x-\cos\Theta)[(2x-\cos\Theta)+\cos\Theta]}{(x^2-2x\cos\Theta+1)^{\frac{3}{2}}}$$
$$- \frac{9x(1-x\cos\Theta)(x-\cos\Theta)^2}{(x^2-2x\cos\Theta+1)^{\frac{5}{2}}}$$
$$= 6 + \frac{3(x-\cos\Theta)+9x}{(x^2-2x\cos\Theta+1)^{1/2}} + \frac{3(1-x\cos\Theta)(x-\cos\Theta)+3x(1-x\cos\Theta)-6x^2(x-\cos\Theta)}{(x^2-2x\cos\Theta+1)^{\frac{3}{2}}}$$
$$- \frac{9x(1-x\cos\Theta)(x-\cos\Theta)^2}{(x^2-2x\cos\Theta+1)^{\frac{5}{2}}}$$

Eq. (6.131)—(6.133)

$$f'(1) = x(1+x^2-2x\cos\Theta)^{\frac{1}{2}} - x + x^2 = (2-2\cos\Theta)^{\frac{1}{2}} = 2\sin\frac{\Theta}{2}$$

$$f''(1) = (1+x^2-2x\cos\Theta)^{\frac{1}{2}} + \frac{x(x-\cos\Theta)}{(1+x^2-2x\cos\Theta)^{\frac{1}{2}}} - 1 + 2x$$
$$= (2-2\cos\Theta)^{\frac{1}{2}} + \frac{(1-\cos\Theta)}{(2-2\cos\Theta)^{\frac{1}{2}}} - 1 + 2 = (2-2\cos\Theta)^{\frac{1}{2}} + \frac{(2-2\cos\Theta)}{2(2-2\cos\Theta)^{\frac{1}{2}}} + 1$$
$$= 1 + (2-2\cos\Theta)^{\frac{1}{2}} + \frac{1}{2}(2-2\cos\Theta)^{\frac{1}{2}} = 1 + \frac{3}{2}(2-2\cos\Theta)^{\frac{1}{2}} = 1 + 3\sin\frac{\Theta}{2}$$

$$f'''(1) = \frac{x}{(1+x^2-2x\cos\Theta)^{\frac{1}{2}}} + \frac{2(x-\cos\Theta)}{(1+x^2-2x\cos\Theta)^{\frac{1}{2}}} - \frac{x(x-\cos\Theta)^2}{(1+x^2-2x\cos\Theta)^{\frac{3}{2}}} + 2$$
$$= \frac{1}{(2-2\cos\Theta)^{\frac{1}{2}}} + \frac{(2-2\cos\Theta)}{(2-2\cos\Theta)^{\frac{1}{2}}} - \frac{(1-\cos\Theta)^2}{(2-2\cos\Theta)^{\frac{3}{2}}} + 2$$
$$= \frac{1}{(2-2\cos\Theta)^{\frac{1}{2}}} + (2-2\cos\Theta)^{\frac{1}{2}} - \frac{(2-2\cos\Theta)^2}{4(2-2\cos\Theta)^{\frac{3}{2}}} + 2$$
$$= \frac{1}{(2-2\cos\Theta)^{\frac{1}{2}}} + (2-2\cos\Theta)^{\frac{1}{2}} - \frac{1}{4}(2-2\cos\Theta)^{\frac{1}{2}} + 2$$
$$= 2 + \frac{1}{(2-2\cos\Theta)^{\frac{1}{2}}} + \frac{3}{4}(2-2\cos\Theta)^{\frac{1}{2}} = 2 + \frac{1}{2\sin\frac{\Theta}{2}} + \frac{3}{2}\sin\frac{\Theta}{2}$$
$$= \frac{1}{2\sin\frac{\Theta}{2}} + 2 + \frac{3}{2}\sin\frac{\Theta}{2} = \frac{1}{2\sin\frac{\Theta}{2}}\left(1 + 4\sin\frac{\Theta}{2} + 3\sin^2\frac{\Theta}{2}\right)$$

Eq. (6.134)—(6.136)

$$3f'(x) = 3x^2 + 3x\sqrt{x^2 - 2x\cos\Theta + 1} - \frac{3x(1 - x\cos\Theta)}{\sqrt{x^2 - 2x\cos\Theta + 1}} = 3 + 3(2 - \cos\Theta)^{\frac{1}{2}} - \frac{3(1 - \cos\Theta)}{(2 - \cos\Theta)^{\frac{1}{2}}} =$$

$$= 3 + 3(2 - \cos\Theta)^{\frac{1}{2}} - \frac{3(2 - 2\cos\Theta)}{2(2 - \cos\Theta)^{\frac{1}{2}}} = 3 + 3(2 - \cos\Theta)^{\frac{1}{2}} - \frac{3}{2}(2 - \cos\Theta)^{\frac{1}{2}}$$

$$= 3 + \frac{3}{2}(2 - \cos\Theta)^{\frac{1}{2}} = 3 + 3\sin\frac{\Theta}{2}$$

$$3f''(1) = 6x + \frac{3x(2x - \cos\Theta)}{\sqrt{x^2 - 2x\cos\Theta + 1}} + \frac{3x(1 - x\cos\Theta)(x - \cos\Theta)}{\left(\sqrt{x^2 - 2x\cos\Theta + 1}\right)^3}$$

$$= 6 + \frac{3(2 - \cos\Theta)}{(2 - \cos\Theta)^{\frac{1}{2}}} + \frac{3(1 - \cos\Theta)^2}{(2 - \cos\Theta)^{\frac{3}{2}}}$$

$$= 6 + \frac{3}{(2 - \cos\Theta)^{\frac{1}{2}}} + \frac{3(1 - \cos\Theta)}{(2 - \cos\Theta)^{\frac{1}{2}}} + \frac{3(1 - \cos\Theta)^2}{(2 - \cos\Theta)^{\frac{3}{2}}}$$

$$= 6 + \frac{3}{(2 - \cos\Theta)^{\frac{1}{2}}} + \frac{3(2 - 2\cos\Theta)}{2(2 - \cos\Theta)^{\frac{1}{2}}} + \frac{3(2 - 2\cos\Theta)^2}{4(2 - \cos\Theta)^{\frac{3}{2}}}$$

$$= 6 + \frac{3}{(2 - \cos\Theta)^{\frac{1}{2}}} + \frac{3(2 - 2\cos\Theta)}{2(2 - \cos\Theta)^{\frac{1}{2}}} + \frac{3(2 - 2\cos\Theta)^2}{4(2 - \cos\Theta)^{\frac{3}{2}}}$$

$$= 6 + \frac{3}{(2 - \cos\Theta)^{\frac{1}{2}}} + \frac{3}{2}(2 - \cos\Theta)^{\frac{1}{2}} + \frac{3}{4}(2 - \cos\Theta)^{\frac{1}{2}}$$

$$= 6 + \frac{3}{(2 - \cos\Theta)^{\frac{1}{2}}} + \frac{9}{4}(2 - \cos\Theta)^{\frac{1}{2}} = 6 + \frac{3}{2\sin\frac{\Theta}{2}} + \frac{9}{2}\sin\frac{\Theta}{2}$$

$$3f'''(1) = 6 + \frac{3(x - \cos\Theta) + 9x}{(x^2 - 2x\cos\Theta + 1)^{1/2}}$$

$$+ \frac{3(1 - x\cos\Theta)(x - \cos\Theta) + 3x(1 - x\cos\Theta) - 6x^2(x - \cos\Theta)}{(x^2 - 2x\cos\Theta + 1)^{\frac{3}{2}}}$$

$$- \frac{9x(1 - x\cos\Theta)(x - \cos\Theta)^2}{(x^2 - 2x\cos\Theta + 1)^{\frac{5}{2}}}$$

$$= 6 + \frac{3(1 - \cos\Theta) + 9}{(2 - 2\cos\Theta)^{1/2}} + \frac{3(1 - \cos\Theta)^2 + 3(1 - \cos\Theta) - 6(1 - \cos\Theta)}{(2 - 2\cos\Theta)^{\frac{3}{2}}}$$

$$- \frac{9(1 - \cos\Theta)^3}{(2 - 2\cos\Theta)^{\frac{5}{2}}}$$

$$= 6 + \frac{3(1-\cos\Theta)+9}{(2-2\cos\Theta)^{\frac{1}{2}}} + \frac{3(1-\cos\Theta)^2 - 3(1-\cos\Theta)}{(2-2\cos\Theta)^{\frac{3}{2}}} - \frac{9(x-\cos\Theta)^3}{(2-2\cos\Theta)^{\frac{5}{2}}}$$

$$= 6 + \frac{3(2-2\cos\Theta)}{2(2-2\cos\Theta)^{\frac{1}{2}}} + \frac{9}{(2-2\cos\Theta)^{\frac{1}{2}}} + \frac{3(2-2\cos\Theta)^2}{4(2-2\cos\Theta)^{\frac{3}{2}}} - \frac{3(2-2\cos\Theta)}{2(2-2\cos\Theta)^{\frac{3}{2}}}$$

$$- \frac{9(2-2\cos\Theta)^3}{8(2-2\cos\Theta)^{\frac{5}{2}}}$$

$$= 6 + \frac{3}{2}(2-2\cos\Theta)^{\frac{1}{2}} + \frac{9}{(2-2\cos\Theta)^{\frac{1}{2}}} + \frac{3}{4}(2-2\cos\Theta)^{\frac{1}{2}} - \frac{3}{2(2-2\cos\Theta)^{\frac{1}{2}}}$$

$$- \frac{9}{8}(2-2\cos\Theta)^{\frac{1}{2}}$$

$$= 6 + \frac{3}{2}(2-2\cos\Theta)^{\frac{1}{2}} + \frac{3}{4}(2-2\cos\Theta)^{\frac{1}{2}} - \frac{9}{8}(2-2\cos\Theta)^{\frac{1}{2}} - \frac{3}{2(2-2\cos\Theta)^{\frac{1}{2}}}$$

$$+ \frac{9}{(2-2\cos\Theta)^{\frac{1}{2}}}$$

$$= 6 + \frac{9}{4}(2-2\cos\Theta)^{\frac{1}{2}} - \frac{9}{8}(2-2\cos\Theta)^{\frac{1}{2}} - \frac{3}{2(2-2\cos\Theta)^{\frac{1}{2}}} + \frac{18}{2(2-2\cos\Theta)^{\frac{1}{2}}}$$

$$= 6 + \frac{9}{8}(2-2\cos\Theta)^{\frac{1}{2}} + \frac{15}{2(2-2\cos\Theta)^{\frac{1}{2}}} = 6 + \frac{9}{4}\sin\frac{\Theta}{2} + \frac{15}{4\sin\frac{\Theta}{2}} = 6 + \frac{9a_r}{8r} + \frac{15r}{2a_r}$$

$$= \frac{15r}{2a_r}\left(1 + \frac{4a_r}{5r} + \frac{3a_r^2}{20r^2}\right)$$

$$= 6 + \frac{3(1-\cos\Theta)+9}{(2-2\cos\Theta)^{1/2}} + \frac{3(1-\cos\Theta)^2 + 3(1-\cos\Theta) - 6(1-\cos\Theta)}{(2-2\cos\Theta)^{\frac{3}{2}}}$$

$$- \frac{9(1-\cos\Theta)^3}{(2-2\cos\Theta)^{\frac{5}{2}}}$$

$$= 6 + \frac{3(1-\cos\Theta)+9}{(2-2\cos\Theta)^{\frac{1}{2}}} + \frac{3(1-\cos\Theta)^2 - 3(1-\cos\Theta)}{(2-2\cos\Theta)^{\frac{3}{2}}} - \frac{9(x-\cos\Theta)^3}{(2-2\cos\Theta)^{\frac{5}{2}}}$$

$$= 6 + \frac{3(2-2\cos\Theta)}{2(2-2\cos\Theta)^{\frac{1}{2}}} + \frac{9}{(2-2\cos\Theta)^{\frac{1}{2}}} + \frac{3(2-2\cos\Theta)^2}{4(2-2\cos\Theta)^{\frac{3}{2}}} - \frac{3(2-2\cos\Theta)}{2(2-2\cos\Theta)^{\frac{3}{2}}}$$

$$- \frac{9(2-2\cos\Theta)^3}{8(2-2\cos\Theta)^{\frac{5}{2}}}$$

$$= 6 + \frac{3}{2}(2-2\cos\Theta)^{\frac{1}{2}} + \frac{9}{(2-2\cos\Theta)^{\frac{1}{2}}} + \frac{3}{4}(2-2\cos\Theta)^{\frac{1}{2}} - \frac{3}{2(2-2\cos\Theta)^{\frac{1}{2}}}$$

$$- \frac{9}{8}(2-2\cos\Theta)^{\frac{1}{2}}$$

$$= 6 + \frac{3}{2}(2-2\cos\Theta)^{\frac{1}{2}} + \frac{3}{4}(2-2\cos\Theta)^{\frac{1}{2}} - \frac{9}{8}(2-2\cos\Theta)^{\frac{1}{2}} - \frac{3}{2(2-2\cos\Theta)^{\frac{1}{2}}}$$

$$+\frac{9}{(2-2\cos\Theta)^{\frac{1}{2}}}$$

$$=6+\frac{9}{4}(2-2\cos\Theta)^{\frac{1}{2}}-\frac{9}{8}(2-2\cos\Theta)^{\frac{1}{2}}-\frac{3}{2(2-2\cos\Theta)^{\frac{1}{2}}}+\frac{18}{2(2-2\cos\Theta)^{\frac{1}{2}}}$$

$$=6+\frac{9}{8}(2-2\cos\Theta)^{\frac{1}{2}}+\frac{15}{2(2-2\cos\Theta)^{\frac{1}{2}}}=6+\frac{9}{4}\sin\frac{\Theta}{2}+\frac{15}{4\sin\frac{\Theta}{2}}=6+\frac{9a_r}{8r}+\frac{15r}{2a_r}$$

$$=\frac{15r}{2a_r}\left(1+\frac{4a_r}{5r}+\frac{3a_r^2}{20r^2}\right)$$

Lengthy Derivations in Sect. 6.4.1

Eq (6.148)

$$\frac{\Delta\rho}{3}[(R-T)^3-(R-T-H')^3]=\frac{\rho}{3}[(R+H)^3-R^3]$$

$$\frac{\Delta\rho}{3}\left[3(R-T)^2H'-3(R-T){H'}^2+H'^3\right]=\frac{\rho}{3}[3R^2H+3RH^2+H^3]$$

$$\left[H'-\frac{{H'}^2}{R-T}+\frac{H'^3}{3(R-T)^2}\right]=\frac{\rho}{\Delta\rho}\frac{R^2}{(R-T)^2}\left[H+\frac{H^2}{R}+\frac{H^3}{3R^2}\right]$$

$$H'=\frac{R^2\rho}{(R-T)^2\Delta\rho}\left[H+\frac{H^2}{R}+\frac{H^3}{3R^2}\right]+\frac{{H'}^2}{R-T}-\frac{H'^3}{3(R-T)^2}$$

$${H'}^2=\frac{R^4\rho^2}{(R-T)^4\Delta\rho^2}\left[H^2+2\frac{H^3}{R}\right]+\frac{2R^2\rho}{(R-T)^3\Delta\rho}HH'^2$$

$${H'}^2=\frac{R^4\rho^2}{(R-T)^4\Delta\rho^2}\left[H^2+2\frac{H^3}{R}\right]+\frac{2R^2\rho}{(R-T)^3\Delta\rho}\frac{R^4\rho^2}{(R-T)^4\Delta\rho^2}H^3$$

$${H'}^2=\frac{R^4\rho^2}{(R-T)^4\Delta\rho^2}\left[H^2+2\frac{H^3}{R}\right]+\frac{2R^6\rho^3}{(R-T)^7\Delta\rho^3}H^3$$

$${H'}^3=\frac{R^6\rho^3}{(R-T)^6\Delta\rho^3}H^3$$

$$H'=\frac{R^2\rho}{(R-T)^2\Delta\rho}\left[H+\frac{H^2}{R}+\frac{H^3}{3R^2}\right]+\frac{R^4\rho^2}{(R-T)^5\Delta\rho^2}\left[H^2+2\frac{H^3}{R}\right]+\frac{2R^6\rho^3}{(R-T)^8\Delta\rho^3}H^3-\frac{R^6\rho^3}{3(R-T)^8\Delta\rho^3}H^3$$

$$H'=\frac{R^2\rho}{(R-T)^2\Delta\rho}\left[H+\frac{H^2}{R}+\frac{H^3}{3R^2}\right]+\frac{R^4\rho^2}{(R-T)^5\Delta\rho^2}\left[H^2+2\frac{H^3}{R}\right]+\frac{5R^6\rho^3}{3(R-T)^8\Delta\rho^3}H^3$$

$$H'=\frac{R^2\rho}{(R-T)^2\Delta\rho}\left[H+\frac{H^2}{R}+\frac{H^3}{3R^2}\right]+\frac{R^4\rho^2}{(R-T)^5\Delta\rho^2}H^2+\frac{2R^3\rho^2}{(R-T)^5\Delta\rho^2}H^3+\frac{5R^6\rho^3}{3(R-T)^8\Delta\rho^3}H^3$$

$$H'=\frac{R^2\rho H}{(R-T)^2\Delta\rho}\left[1+\frac{H}{R}\left(1+\frac{R^3\rho}{(R-T)^3\Delta\rho}\right)+\frac{H^2}{3R^2}\left(1+\frac{6R^3\rho}{(R-T)^3\Delta\rho}+\frac{5R^6\rho^2}{(R-T)^6\Delta\rho^2}\right)\right]$$

Lengthy Derivations in Sect. 6.5.4

Eq (6.230)

$$
\begin{aligned}
f(x,\psi) &= (x + 3\,cos\,\psi)(1 + x^2 - 2x\,cos\,\psi)^{1/2} \\
&\quad + (3\,cos^2\,\psi - 1)\,ln\left[(1 + x^2 - 2x\,cos\,\psi)^{1/2} + x - cos\,\psi\right]
\end{aligned}
$$

$$
\begin{aligned}
\frac{\partial}{\partial\psi} f(x,\psi) &= -3\sin\psi\,(1 + x^2 - 2x\,cos\,\psi)^{\frac{1}{2}} + \frac{x\sin\psi\,(x + 3\,cos\,\psi)}{(1 + x^2 - 2x\,cos\,\psi)^{\frac{1}{2}}} \\
&\quad + \frac{(3\,cos^2\,\psi - 1)\left[\dfrac{x\sin\psi}{(1 + x^2 - 2x\,cos\,\psi)^{\frac{1}{2}}} + \sin\psi\right]}{\left[(1 + x^2 - 2x\,cos\,\psi)^{\frac{1}{2}} + x - cos\,\psi\right]} \\
&\quad - 6\sin\psi\cos\psi\,ln\left[(1 + x^2 - 2x\,cos\,\psi)^{1/2} + x - cos\,\psi\right] \\
&= -3\sin\psi\,(1 + x^2 - 2x\,cos\,\psi)^{\frac{1}{2}} + \frac{x\sin\psi\,(x + 3\,cos\,\psi)}{(1 + x^2 - 2x\,cos\,\psi)^{\frac{1}{2}}} \\
&\quad + \frac{\sin\psi\,(3\,cos^2\,\psi - 1)\left[(1 + x^2 - 2x\,cos\,\psi)^{\frac{1}{2}} + x\right]}{(1 + x^2 - 2x\,cos\,\psi)^{\frac{1}{2}}\left[(1 + x^2 - 2x\,cos\,\psi)^{\frac{1}{2}} + x - cos\,\psi\right]} \\
&\quad - 6\sin\psi\cos\psi\,ln\left[(1 + x^2 - 2x\,cos\,\psi)^{1/2} + x - cos\,\psi\right] \\
&= -3\sin\psi\,(1 + x^2 - 2x\,cos\,\psi)^{\frac{1}{2}} + \frac{x\sin\psi\,(x + 3\,cos\,\psi)}{(1 + x^2 - 2x\,cos\,\psi)^{\frac{1}{2}}} \\
&\quad + \frac{\sin\psi\,(3\,cos^2\,\psi - 1)}{(1 + x^2 - 2x\,cos\,\psi)^{\frac{1}{2}} + x - cos\,\psi} \\
&\quad + \frac{x\sin\psi\,(3\,cos^2\,\psi - 1)}{(1 + x^2 - 2x\,cos\,\psi)^{\frac{1}{2}}\left[(1 + x^2 - 2x\,cos\,\psi)^{\frac{1}{2}} + x - cos\,\psi\right]} \\
&\quad - 6\sin\psi\cos\psi\,ln\left[(1 + x^2 - 2x\,cos\,\psi)^{1/2} + x - cos\,\psi\right]
\end{aligned}
$$